Circular Motion Revision Sheet

Level 1 Sheet

Note: Revise Class Notes + Do Last 5 Years of PYQ + These Sheets of Level 1 to 3.

1. A man carrying a monkey on his shoulder does cycling smoothly on a circular track of radius 9 m and completes 120 revolutions in 3 minutes. The magnitude of centripetal acceleration of monkey is (in m/s^2) :

(A) $57600\pi^2$ ms^{-2} (B) Zero

(C) $4\pi^2$ ms^{-2} (D) $16\pi^2$ ms^{-2}

2. A particle is moving with constant speed in a circular path. When the particle turns by an angle $90°$, the ratio of instantaneous velocity to its average velocity is $\pi : x\sqrt{2}$. The value of x will be

(A) 2 (B) 5

(C) 1 (D) 7

3. A stone of mass 1 kg is tied to end of a massless string of length 1 m. If the breaking tension of the string is 400 N, then maximum linear velocity, the stone can have without breaking the string, while rotating in horizontal plane, is:

(A) 20 $m\,s^{-1}$ (B) 40 $m\,s^{-1}$

(C) 400 $m\,s^{-1}$ (D) 10 $m\,s^{-1}$

4. An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at $x = +2$ m, its velocity is $-4\hat{j}$ $m\,s^{-1}$. The object's velocity (v) and acceleration (a) at $x = -2$ m will be

(A) $v = 4\hat{i}$ $m\,s^{-1}$, $a = 8\hat{j}$ $m\,s^{-2}$ (B) $v = 4\hat{j}$ $m\,s^{-1}$, $a = 8\hat{i}$ $m\,s^{-2}$

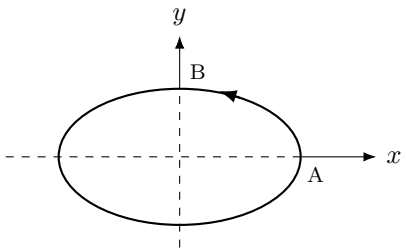
(C) $v = -4\hat{j}$ $m\,s^{-1}$, $a = 8\hat{i}$ $m\,s^{-2}$ (D) $v = -4\hat{i}$ $m\,s^{-1}$, $a = -8\hat{j}$ $m\,s^{-2}$

5. A car is moving on circular path of radius 100 m such that its speed is increasing at the rate of 5 m/s^2. At $t = 0$ it starts from rest, what is the radial acceleration of car at the instant it completed one round trip?

(1) 40π m/s^2 (2) 10π m/s^2 (3) 20π m/s^2 (4) 15π m/s^2

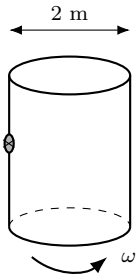
6. A particle performs uniform circular motion. At some instant of time its acceleration is $\vec{a} = x\hat{i} + 3\hat{j}$ and velocity is $\vec{v} = 3\hat{i} - 2\hat{j}$. Find the value of x (in m/s^2) at that time. All quantities are in S.I. units.

7. A particle is moving along an elliptical path with a constant speed. As it moves from A to B, magnitude of its acceleration :

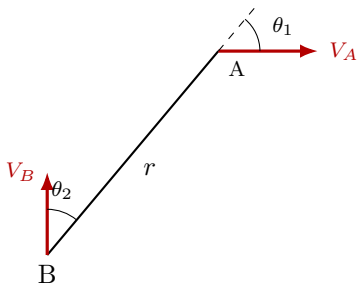


(1) continuously increases
(2) continuously decreases
(3) Remains constant
(4) first increases and then decreases

8. An insect of mass $m = 3$ kg is inside a vertical drum of radius 0.2 m that is rotating with an angular velocity of 5 rad/sec. The insect does not fall off, then the minimum co-efficient of friction required is:

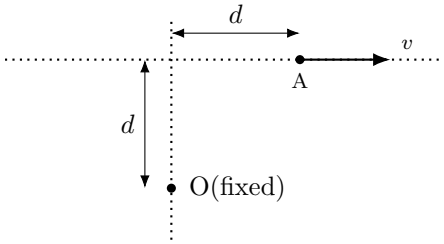


9. Find the angular velocity of A with respect to B in the figure given below:



(Revise relative omega acche se from class notes, solve 3 numericals from class notes itself)

10. Find angular velocity of A with respect to O at the instant shown in the figure.



11. A particle begins to move with a tangential acceleration of constant magnitude 0.6 m/s^2 in a circular path. If it slips when its total acceleration becomes 1 m/s^2, Find the angle through which it would have turned before it starts to slip.

12. A particle moves in a circle of radius 2.0 cm at a speed given by $v = 4t$, where v is in cm/s and t is in seconds.
(a) Find the tangential acceleration at $t = 1$ s.
(b) Find total acceleration at $t = 1$ s.

13. A block of mass 2 kg is tied to a string of length 2 m, the other end of which is fixed. The block is moved on a smooth horizontal table with constant speed 5 m/s. Find the tension in the string.

(Note- Prepare Radius of curvature, Conical Pendulum and circular motion kinematics from class Notes, Practice 5 Questions on each)

Keep Going!

"Believe you can and you're halfway there."

— Theodore Roosevelt